FINAL PROJECT UPDATE / 03 NOVEMBER 2026

CSTBS Version 5.4

Stadium Operations & Ticket Experience

Release objective: Give stadium administrators clear control of entrances and attendance; give students a separate ticket dashboard; make camera-based QR entry usable across more phones and browsers.

What changed

Area	Delivered in V5.4
Stadium admin dashboard	Event status, reservation/attendance numbers, prominent Open Entry / Close Entry, gate and seat controls.
Student dashboard	Separate /dashboard for upcoming events, own tickets, classmates-booked passes and student account.
Named entrance gates	Gate entity per event, individual open/close, section-to-gate assignments, gate details on tickets.
Cross-browser QR capture	Native live browser scanner where supported; server-side OpenCV decoding of uploaded mobile QR photographs.
Admission verdicts	Valid ticket waiting for opening, wrong gate, closed gate, invalid, cancelled, used and ready-to-admit results.
Ticket presentation	Date, time, venue, gate, section, row, seat, attendee, Student ID, ticket number, free admission and signed QR.
Uniqueness & UI	One active ticket per student per event; user-friendly role navigation, responsive cards, no version tagline in footer.

Admission is free for all events. No payment provider or checkout charge is introduced.

ENTRY OPERATIONS / CONTROL FLOW

Scan safely. Admit once.

Gate workflow

01 Assign entrances

Admin opens Stadium Control, visits Manage Gates and maps the event’s seating sections to Gate 1, Gate 2, etc.

02 Open entry

Admin uses Open entry (all gates) or opens individual gates. Admission remains closed by default for new/upgraded events.

03 Select and scan

Staff selects their physical gate, uses live camera over HTTPS/localhost or uploads a QR photo. No QR scan itself grants entry.

04 Verify identity

CSTBS checks ticket signature, attendee, booking, event entrance status and the assigned gate. Staff compares university ID.

05 Confirm once

A protected POST records a checked-in status, operator and entrance; a second attempt cannot admit the same pass.

How valid tickets differ from invalid tickets

Screen result	Operator action
VALID TICKET - ADMIT STUDENT	Check university ID and confirm admission once.
VALID TICKET - ENTRY NOT OPEN	Ask admin to open entry for this event; do not reject QR as fake.
VALID TICKET - WRONG / CLOSED GATE	Redirect to the assigned entrance or ask admin to open that gate.
CANCELLED / ALREADY CHECKED IN / INVALID	Do not admit; check ticket status and record an incident if appropriate.

QR privacy: the pass contains a signed opaque token, not the student’s personal details. It is verified against the database on every scan. Uploaded photographs are processed on the local server and not saved.

MIGRATION / RELEASE READINESS

GitHub-safe delivery

Database and compatibility

V5.4 adds a Gate model, a seat-to-gate reference and a checked-in-gate audit field. The idempotent SQLite migration assigns each legacy event a Gate 1, preserves existing accounts, seat IDs, bookings, checked-in states and QR signing data, and rejects gates from other events. A pre-upgrade database snapshot is created for older schemas. Do not replace the live database when pushing code to GitHub.

Installation and smoke test (Windows PowerShell)

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m flask --app run.py init-db
python -m flask --app run.py seed-events
python -m unittest discover -s tests -v
python run.py
```

Offline validation completed

Python syntax/compilation, 29 Jinja templates, three JavaScript files, idempotent V5.4 SQLite upgrade, cross-event gate protection and an OpenCV QR-photo decode smoke test passed.

Still required before campus use

Flask HTTP integration tests, QR scanning on real staff phones over the campus HTTPS origin, gate open/close timing, concurrent scanner duplicate-entry tests, SMTP, access controls and a full backup/restore drill were not executed in this isolated environment. Do not treat this release as production certified until they pass.

GitHub release information

Field	Value
Tag	v5.4
Release title	CSTBS v5.4 - Stadium Operations & Ticket Experience
Suggested commit	feat: improve CSTBS v5.4 stadium gates and ticket dashboards
CI protection	Read-only GitHub Actions tests on push / pull request; no automatic deployment.

Full source, migration, tests, operational guides and release text are packaged in CSTBS-v5.4.zip.